

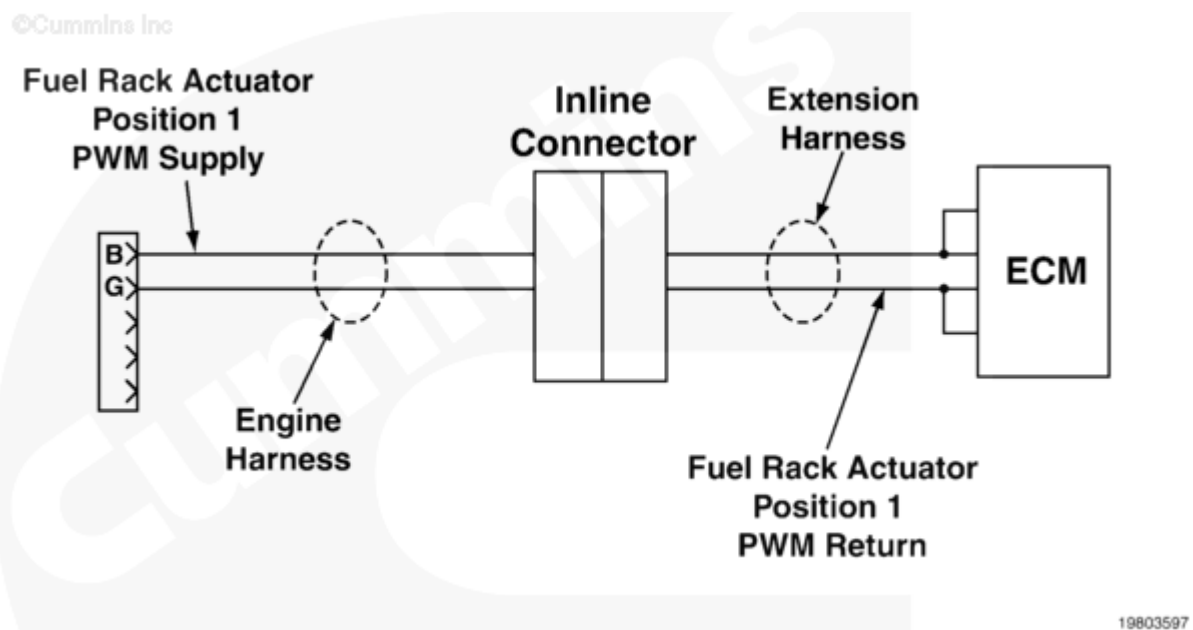
## Fault Code: 171

### Fuel Rack Actuator Position 1 Circuit - Data Incorrect (Left Bank)

 Printable Version

#### Overview

| Codes                                                                   | Reason                                                              | Effect                                          |
|-------------------------------------------------------------------------|---------------------------------------------------------------------|-------------------------------------------------|
| Fault Code: 171<br><br>PID(P):<br>SPN:<br>FMI:<br>Lamp: Warning<br>SRT: | Fuel rack actuator position 1 circuit - data incorrect (left bank). | Performance can be sluggish or slow to respond. |



Fuel Rack Actuator Position 1 Circuit

#### Circuit Description

The fuel rack position sensor circuit's measured, reference and common signals are used by the ECM to verify that the rack has been properly positioned by the rack actuator. This fault code indicates that either the rack is mechanically stuck, or the rack position sensor data is incorrect. This is accomplished as follows:

The ECM sends out a fixed frequency and voltage signal on the reference signal line to the fuel pump. The ECM also sends out a fixed frequency but variable voltage signal that is 180 degrees out of phase from the reference signal from the measured signal line to the fuel pump. The amplitude of the measured signal voltage is measured in the ECM and represents the rack position of the fuel pump.

Circuitry inside the fuel pump compares the difference in voltage between the reference signal input and the measured signal input to the pump and sends a signal back to the ECM that is the difference of these two signals. The difference voltage signal is output on the common signal line back to the ECM. The amplitude of the common signal is dependent on the position of the rack inside the fuel pump. Circuitry inside the ECM determines if the common signal is below a certain threshold. If it is, this means that the rack position of the fuel pump is at the commanded position. The voltage value of the measured signal line is then measured by the ECM and represents the rack position.

## Component Location

The fuel pump rack, rack actuator, and rack position sensor are all located within the fuel pump. This fault code requires engine speed greater than zero before the fault will go active.

## Shoptalk

The possible failure modes are rack mechanically stuck, rack position sensor stuck in range, rack position sensor out of calibration, or bad fuel pump.

This fault code requires engine speed greater than zero before the fault will go active.

Fault Code 171 can go active after Fault Code 168 is corrected. If this is the case, key off and Fault Code 171 will go inactive and can be cleared

Refer to Troubleshooting Fault Code t05-171 (</qs3/pubsys2/xml/en/procedures/01/01-t05-171.html>)